CHAPTER 4

THE D- AND F-BLOCK ELEMENTS

CHAPTER ESSENCE AT A GLANCE:
Electronic configurations, variable oxidation states, catalytic and magnetic properties of d-block transition metals, and lanthanoid contraction in f-block.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Transition Elements & Electronic Structure:** Defined by incompletely filled d-orbitals in their ground state or common oxidation states; elements like zinc, cadmium, and mercury are excluded from typical transition series due to completely filled d-subshells in both elemental and oxidized forms.

• **Variable Oxidation States & Catalytic Action:** The comparable energy levels of valence ns and inner (n-1)d orbitals permit flexible electron loss, yielding multiple stable oxidation states and allowing transition metal compounds to form catalytic intermediates and interstitial alloys.

• **Crystal Field Color & Magnetic Moments:** Ligand field interactions split degenerate d-orbitals into distinct energy levels, permitting d-to-d electronic transitions that absorb specific visible wavelengths and impart complementary colors; paramagnetic behavior scales with the number of unpaired d-electrons.

• **Potassium Dichromate & Permanganate Reagents:** Powerful industrial and laboratory oxidizing agents prepared from chromite and pyrolusite ores; their oxidizing strength relies on reduction to lower valent chromium and manganese species in acidic media.

ESSENTIAL FORMULAS & GOVERNING EQUATIONS

$$\text{Configuration: } (n-1)d^{1-10}ns^{1-2} \quad (\text{e.g. Cr: } 3d^5 4s^1,\ \text{Cu: } 3d^{10}4s^1)$$

General valence electronic configuration of transition elements

$$\mu = \sqrt{n(n+2)}\ \text{BM} \quad (n = \text{number of unpaired electrons})$$

Spin-only magnetic moment formula in Bohr Magnetons (BM)

$$Cr_2O_7^{2-} + 14H^+ + 6e^- \rightarrow 2Cr^{3+} + 7H_2O \quad (E^\circ = +1.33\ \text{V})$$

Potassium dichromate reduction half-reaction in acidic medium

$$MnO_4^- + 8H^+ + 5e^- \rightarrow Mn^{2+} + 4H_2O \quad (E^\circ = +1.51\ \text{V})$$

Potassium permanganate reduction half-reaction in acidic medium

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** Magnetic moment μ = sqrt(n(n + 2)) BM, where n is number of UNPAIRED electrons.

• **Alert:** [Valuation Rubric] Write ground state electronic configuration of transition metal and ion .

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Why are transition elements called d-block elements? Write their general electronic configuration. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Configuration]** General electronic configuration: $(n-1)d^{1-10}$ ns^{1-2}. [1 Mark]

• **[Step 2: Zn, Cd, Hg Exclusion]** A transition metal has partially filled d-orbitals in ground state or any stable oxidation state. Zn ($[Ar]3d^{10}$ $4s^2$), Cd ($[Kr]4d^{10}$ $5s^2$), Hg ($[Xe]4f^{14}$ $5d^{10}$ $6s^2$) have completely filled d-subshell (d^{10}) in elemental state and +2 oxidation state (Zn^{2+} $3d^{10}$). Hence they are not typical transition metals. [2 Marks]

• **Key Exam Result:** $(n-1)d^{1-10}$ ns^{1-2}; Zn/Cd/Hg excluded due to full d^{10} subshell

2. Why do transition metals exhibit variable oxidation states? Which 3d element has the highest oxidation state? [Score: 2 Marks]

Answer & Key Valuation Steps:

• **[Step 1: High Melting Points]** Due to strong interatomic metallic bonding involving both ns and (n-1)d electrons. [1 Mark]

• **[Step 2: Variable Oxidation States]** Because the energy difference between ns and (n-1)d orbitals is very small, allowing both to participate in chemical bonding (e.g. Mn shows +2 to +7). [1 Mark]

• **[Step 3: Catalytic Properties]** Due to: 1. Ability to adopt multiple oxidation states. 2. Provision of large surface area and vacant d-orbitals to form intermediate complexes. [1 Mark]

• **Key Exam Result:** High MP (metallic bonding), Variable OS (small ΔE of ns divided by (n-1)d), Catalytic (variable OS & vacant d-orbitals)

3. Why do transition metal compounds show catalytic properties and form interstitial compounds? [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Colored Ions]** In presence of ligands, d-orbitals split into two sets (t {2g} and e g). Electrons absorb visible light and transition from lower to higher d-orbital (d-d transition). The transmitted complementary light gives color. [1.5 Marks]

• **[Step 2: Complex Formation]** Form complexes easily due to: 1. Small ionic size and high nuclear charge (high charge or radius ratio). 2. Availability of vacant d-orbitals of appropriate energy to accept lone pairs from ligands. [1.5 Marks]

• **Key Exam Result:** Color due to d-d transition; Complexes due to vacant d-orbitals and high charge density

4. Describe the preparation of potassium dichromate (K2Cr2O7) from chromite ore with balanced equations. [Score: 4 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Roasting with Na_2CO_3]** $4FeCr_2O_4 + 8Na_2CO_3 + 7O_2 \rightarrow 8Na_2CrO_4$ (yellow) $+ 2Fe_2O_3 + 8CO_2$. [1.5 Marks]

• **[Step 2: Acidification & Conversion]** $2Na_2CrO_4 + H_2SO_4 \rightarrow Na_2Cr_2O_7$ (orange) $+ Na_2SO_4 + H_2O$. $Na_2Cr_2O_7 + 2KCl \rightarrow K_2Cr_2O_7$ (orange crystals) + 2NaCl (less soluble, crystallizes out). [1.5 Marks]

• **Key Exam Result:** Chromite ore $\rightarrow Na_2CrO_4 \rightarrow Na_2Cr_2O_7 \rightarrow K_2Cr_2O_7$

5. Describe the preparation of potassium permanganate (KMnO4) from pyrolusite ore with balanced equations. [Score: 4 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Fusion with KOH]** $2MnO_2 + 4KOH + O_2 \rightarrow 2K_2MnO_4$ (potassium manganate, dark green) $+ 2H_2O$. [1.5 Marks]

• **[Step 2: Electrolytic or Acidic Oxidation]** $3MnO_4^{2-} + 4H^+ \rightarrow 2MnO_4^-$ (permanganate, deep purple) $+ MnO_2 + 2H_2O$. Industrial: Electrolytic oxidation of alkaline manganate: $MnO_4^{2-} \rightarrow MnO_4^- + e^-$. [1.5 Marks]

• **Key Exam Result:** $MnO_2 \rightarrow K_2MnO_4$ (green) $\rightarrow KMnO_4$ (purple)

6. Write balanced ionic equations for oxidizing action of acidified KMnO4 on (a) Fe^{2+}, (b) oxalate ion $C2O4^{2-}$, (c) I^-. [Score: 3 Marks]

Answer & Key Valuation Steps:

- **[Step 1: With Fe^{2+}]** MnO_4^- + $5Fe^{2+}$ + $8H^+$ --> Mn^{2+} + $5Fe^{3+}$ + $4H_2O$. [1.5 Marks]
- **[Step 2: With Oxalate $C_2O_4^{2-}$]** $2MnO_4^-$ + $5C_2O_4^{2-}$ + $16H^+$ --> $2Mn^{2+}$ + $10CO_2$ + $8H_2O$. [1.5 Marks]
- **Key Exam Result:** MnO_4^- oxidizes Fe^{2+} to Fe^{3+} and $C_2O_4^{2-}$ to CO_2

7. Write balanced ionic equations for oxidizing action of acidified K2Cr2O7 on (a) Fe^{2+}, (b) I^-, (c) H2S. [Score: 3 Marks]

Answer & Key Valuation Steps:

- **[Step 1: With Fe^{2+}]** $Cr_2O_7^{2-}$ + $6Fe^{2+}$ + $14H^+$ --> $2Cr^{3+}$ + $6Fe^{3+}$ + $7H_2O$. [1.5 Marks]
- **[Step 2: With Iodide I^-]** $Cr_2O_7^{2-}$ + $6I^-$ + $14H^+$ --> $2Cr^{3+}$ + $3I_2$ + $7H_2O$. [1.5 Marks]
- **Key Exam Result:** $Cr_2O_7^{2-}$ oxidizes Fe^{2+} to Fe^{3+} and I^- to I_2